RESOURCE OPTIMIZATION SYSTEM FOR COLLEGE

Mini Project report submitted in partial fulfilment for the requirement of the Degree of

MASTER OF SCIENCE IN SOFTWARE SYSTEMS

MANJUNATHNAN.M

23MSS026

Under the Guidance of

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MARCH 2026



Sri Krishna Arts and Science College

Reaccredited by NAAC with 'A' Grade

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Kuniamuthur, Coimbatore -641008

DECLARATION

I hereby declare that the Mini Project report entitled "RESOURCE OPTIMIZATION SYSTEM FOR COLLEGE" submitted in partial fulfillment of the requirements for the award of degree of Master of Science in Software Systems is an original work and it has not previously formed the basis for the award of any Degree, Diploma, Associateship, Fellowship or similar titles to any other university or body during the period of my study.

Place: Coimbatore

Date:



Signature of the Candidate

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CERTIFICATE

This is to certify that the Mini Project report entitled "RESOURCE OPTIMIZATION SYSTEM FOR COLLEGE" submitted in partial fulfillment of requirements for the degree of Master of Science in Software Systems is a record of Bonafide work carried out by MANJUNATHAN.M 23MSS026 and that no part of this has been submitted for the award of any other degree or diploma and the work has not been published in popular journal or magazine.

GUIDE HOD DEAN

This Project Report is submitted for the Viva Voce examination conducted on _____at Sri Krishna Arts and Science College.

Internal Examiner External Examiner

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Abstract

The Resource Optimization System for College is a software-based solution developed to efficiently manage and allocate institutional resources such as classrooms, laboratories, faculty members, equipment, and academic schedules. In many colleges, manual planning and traditional management methods often result in resource underutilization, scheduling conflicts, and increased administrative workload. This system is designed to overcome these challenges through an automated and systematic approach to resource allocation.

The proposed system analyzes resource availability, academic requirements, and predefined constraints to generate optimized allocation plans. It ensures effective timetable generation, proper faculty assignment, and optimal room utilization while minimizing clashes and idle time. The system also provides real-time monitoring and reporting features to track resource usage and performance.

By implementing optimization techniques and centralized data management, the system enhances operational efficiency, reduces errors, and improves transparency in resource distribution. It assists administrators in making informed decisions through analytical reports and utilization summaries.

Overall, the Resource Optimization System improves institutional productivity, reduces operational costs, and supports better academic planning, thereby contributing to efficient college management and improved educational outcomes.

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